ABDUR RAFAY

AI/ML Engineer | Data Analyst | Software Engineer

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SUMMARY

Software Engineering graduate (BS, University of Karachi) focused on AI/ML, NLP, LLM applications and data analysis. Built and deployed end-to-end projects including a RAG assistant, an ML-based email threat analyzer and a competitive intelligence platform using Python, FastAPI and React. Seeking an internship or entry-level role in AI/ML engineering or data analytics.

SKILLS AND TOOLS

Languages: *Python, SQL, JavaScript, TypeScript, HTML, CSS*
AI/ML: *Machine Learning, Deep Learning, NLP, LLMs, RAG, CNN, LSTM, XGBoost, DistilBERT, Sentence-Transformers, FAISS, TensorFlow, Scikit-learn, OCR*
Data: *Pandas, NumPy, Matplotlib, EDA, Data Preprocessing, ETL, Data Visualization, Power BI, Excel, Web Scraping (BeautifulSoup, Playwright)*
Backend and Web: *FastAPI, REST APIs, SQLAlchemy, PostgreSQL, SQLite, React, Tailwind CSS, Node.js*
Tools: *Git, GitHub, Docker, Redis, Vercel, Jira, Shopify, WordPress, BigCommerce*

PROJECTS

RAG University Assistant (Python, FastAPI, React, TypeScript, PostgreSQL, Groq LLM)

- Built a retrieval-augmented generation assistant that answers questions on university policies and handbooks, with streamed responses, source citations and confidence scores.
- Implemented PDF, DOCX and TXT ingestion with chunking and full-text retrieval; delivered two API backends (Express and FastAPI) with an admin analytics dashboard.

Email Threat Analyzer (Python, FastAPI, React, XGBoost, DistilBERT, Docker)

- Developed a three-layer ML pipeline (XGBoost, DistilBERT, TF-IDF) to classify phishing and malicious emails, with IOC graph visualization and PDF reports.

Signal-Sentry: Competitive Intelligence Platform (Python, FastAPI, Redis, Playwright, Docker)

- Built a monitoring pipeline (scrape, snapshot, diff, classify, alert, digest) that tracks competitor pricing and product changes, with background job queues and a live dashboard.

Resume-Buildr: AI Resume Screener (Python, FastAPI, React, FAISS, Sentence-Transformers)

- Created an ATS-style screener that scores resumes against job descriptions using skill match, semantic similarity, experience and education, with candidate ranking.

FYP Cybersecurity Virtual Assistant (Python, FastAPI, Llama.cpp, OCR)

- Built a chatbot UI with a quantized LLM and OCR to read cyberthreat screenshots and generate guidance for the problem.

Other Projects: Cancer Risk Analysis (Pandas, Scikit-learn); Coffee Shop Performance Dashboard (Power BI, Excel); Cat/Dog Image Classification (CNN, TensorFlow)

PROFESSIONAL EXPERIENCE

CODUP **Karachi (On-site)**
Intern Business Analyst **Aug 2025 - Nov 2025**

- Wrote Jira tickets, top-level approach documents and a Requirement Analysis Document for a new lead project.
- Attended client meetings to gather business requirements and communicated them to cross-functional teams.
- Configured stores on WordPress/WooCommerce, Shopify Plus (B2B and B2C) and BigCommerce to evaluate ecommerce features.

EDUCATION

Karachi University (UBIT) **Karachi, Sindh**
Bachelor of Science - BS (SE), Software Engineering **2021-2025**

CERTIFICATIONS

AI, Machine Learning, Deep Learning and Communication (NAVTTC Program) **Feb-May 2026**
SQL for Data Analytics (SimpliLearn) **May 2025**
Business Analytics using Excel (SimpliLearn) **Jan 2024**
Intro to Machine Learning (self-paced) **Feb 2024**
Python In-depth Course for Beginners (Udemy) **May 2021**

ACHIEVEMENTS

- **5th position** in IEEE YESIST12 Final Year Project Ideas Competition (Cyber Security Virtual Assistant).